

Chapter 69 — Male Infertility

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EPIDEMIOLOGY

- Infertility affects approximately **15% of couples**, leaving nearly **1 in 6 childless** (World Health Organization [WHO]).
- Despite historical bias in the sex split, males and females contribute **an equal number of faulty gametes** — so evaluating the male matters.
- Male care is largely delivered **outside reimbursement systems**, so metrics are unreliable. The Society for Assisted Reproductive Technology (SART), the professional group of the American Society for Reproductive Medicine (ASRM), compels in vitro fertilisation (IVF) clinics to report outcomes — but this sees the male contribution **through the lens of female assisted reproduction**, skewing estimates.
- National Survey of Family Growth (NSFG): **a substantial minority of males in infertile couples are never evaluated**, supporting underreporting of male factor; of sexually active males who ever seek care for difficulty conceiving, only a minority report a clinician-diagnosed male reproductive problem.
- Urologic Diseases in America (UDA) Project: ambulatory surgery for conditions associated with male infertility is used most at **25–34 years**, falling with each older band; younger males account for **more than half** of cases, yet nearly **1 in 11 cases** occurs in the fifth decade or later. Use is lower in the West than in the Northeast and Midwest of the United States.
- Economics: direct expenditure on primary male infertility is a **gross underestimate** because much care escapes traditional databases; counting the female partner's assisted reproduction, the true cost is **orders of magnitude larger**; males pursuing fertility care spend a substantial fraction of annual income on it.
- The primary assay, the **semen analysis, is a poor predictor** — every parameter has a low area under the receiver operating characteristic (ROC) curve — so men with some sperm are wrongly judged fertile and omitted from tallies.

Concept — *the male contribution is systematically undercounted, because every data source is filtered through systems built for the female partner and through an assay that cannot separate the fertile from the infertile.*

HISTORY

- Male gamete production needs **orchestration of endocrine, immune and neural systems**, intact anatomy, sequenced gene expression and chromosomal events, and normal embryological and postnatal development — so **myriad disparate conditions** cause dysfunction.
- Approach the history **methodically**; a **pre-visit form** completed at home or in the waiting area improves efficiency.

TABLE 69.1 DISTRIBUTION OF FINAL DIAGNOSES FROM A MALE INFERTILITY CLINIC

Frequency tier	Diagnoses (in descending order)
Dominant (together over half)	Idiopathic; varicocele
Common	Obstruction; normal (female factor)
Uncommon (each a few per cent or less)	Cryptorchidism; immunological; ejaculatory failure; endocrinological; drug or radiation; genetic; testicular failure
Rare (each under 1%)	Sexual dysfunction; pyospermia; cancer; systemic disease; infection; torsion; ultrastructural

From Sigman, Lipshultz and Howards, Office evaluation of the subfertile male, in *Infertility in the Male*, 4th ed, 2009.

- **What the table is for:** the breadth of final diagnoses in male infertility.
- **Pattern:** idiopathic is the largest single category; **varicocele is the commonest identifiable cause**, then obstruction.
- **Trap:** the percentages vary widely with how each condition is assessed — the authors call the table an indictment of how poorly male reproductive data are collected; do not treat the figures as population truths.
- **Two patients are required** for a positive outcome, so the diagnostic approach must be **probabilistic**.
- With optimally timed intercourse, **most conceptions occur within the first three to six cycles** and nearly all couples who will conceive have done so by twelve; a minority not conceiving by six cycles still do so.
- Couples seeking help one or two months after stopping contraception should be **counselled to keep trying**; it is **reasonable to begin evaluation after 6 months**, accepting that some will conceive shortly afterwards. Describe each month as **rolling a die or flipping a coin**.
- **Ask about intercourse frequency.** Semen parameters **peak after 1–2 days of abstinence** and then decline; prolonged abstinence to "accumulate" sperm **worsens** quality.
- For **conception: daily intercourse around ovulation** (for semen analysis, see Evaluation of Semen).
- **Female partner's age** is critical to reproductive potential and planning, especially with assisted reproduction; fecundity **declines precipitously after 35**, and faster in some regions — assess her age in the context of her locale.
- **Male age** affects bulk parameters and sperm DNA packaging only to a limited degree, and males father children well into later life. However, the ASRM and American Urological Association (AUA) joint guideline (2021) directs clinicians to advise couples with **advanced paternal age (≥40 years)** of increased risk of adverse offspring outcomes — **bipolar disorder, schizophrenia, autism, attention-deficit disorder** and poorer cognitive performance.
- Ask whether the man and partner have **conceived before**, together or with others, and the offspring's ages: **proven post-pubertal fertility eliminates many congenital concerns**.
- **Diabetes mellitus (DM), multiple sclerosis (MS) and spinal cord injury (SCI)** disrupt coordinated ejaculation.
- Cancer impairs spermatogenesis **even before chemotherapy**, especially testis cancer; **azoospermia may reveal testis cancer** — consider it in any man with no sperm.
- **Transurethral resection of the prostate (TURP)** and minimally invasive prostatic therapies cause ejaculatory dysfunction; **retroperitoneal lymph node dissection** causes retrograde ejaculation of varying degree depending on template and context.
- **Herniorrhaphy** can obstruct the vas; **mesh** incites a dense foreign-body inflammatory reaction that may entrap the vas even when not adjacent to it. Infertility from vasal occlusion alone needs **both vasa occluded** (rare) — more often **one occluded vas plus contralateral spermatogenic dysfunction**.
- Without a form, use the mnemonic **TICS**: Toxins, Infectious/inflammatory disease, Childhood history, Sexual history.

Spermatotoxicity

Many medications, prescribed and over the counter, impair spermatogenesis, mature sperm function or delivery.

Endocrine Modulators

- Drugs shift the **oestrogen-to-androgen ratio** by: oestrogen-like structure; ↑ oestrogen synthesis; ↑ aromatase; displacement of steroids from sex hormone-binding globulin (SHBG); ↓ testosterone (T)

synthesis; competitive and non-competitive steroid-receptor binding; ↓ adrenal steroid synthesis; hyperprolactinaemia.

ENDOCRINE-MODULATING AGENTS WARRANTING INQUIRY

Class	Examples	Note
Antiandrogens	Bicalutamide, flutamide, nilutamide	Receptor blockade
Antihypertensive	Spironolactone	Antiandrogenic
Antiretroviral protease inhibitors	Indinavir	
Nucleoside reverse transcriptase inhibitors	Stavudine	
Corticosteroids	—	Especially in adolescence
Oestrogens	Exogenous oestrogen	
5α-reductase inhibitors (5ARIs)	Finasteride, dutasteride	Limited suppression, if any
Exogenous testosterone / androgenic steroids	—	Suppress spermatogenesis

- **5ARIs:** anecdotal reports and a prospective database review show dramatic improvement after stopping (even the low alopecia dose), but **semen inter-assay variability** means this may be chance.
- **Exogenous T:** aromatised to estradiol (E2) → **inhibits pituitary luteinising hormone (LH)** → ↓ intratesticular T → ↓ spermatogenesis. Other androgenic agents act similarly.
- T and androgens have been studied as **male contraceptives since the 1970s**; with ≤2 years' exposure suppression is **reversible**, sperm returning about **4 months or more** after stopping; recovery after longer use is **unknown**.

Recreational Drugs

COMPARISON — RECREATIONAL EXPOSURES AND MALE REPRODUCTION

Agent	Effect	Strength of evidence
Cannabis	Dose- and duration-dependent ↓ plasma T; non-linear ↓ concentration and total count with a threshold above once weekly ; chronic THC edibles in monkeys → dose-response testicular atrophy, ↑ gonadotropins, ↓ sex steroids	Conflicting, most studies adverse
Alcohol	Chronic intake ↓ androgens and sperm parameters; heavy intake ↑ aromatisation of T to E2; even moderate intake may ↓ intracytoplasmic sperm injection (ICSI) outcomes	More robust
Cigarettes	Dose-dependent deterioration of semen; ↑ seminal oxidative stress; ↓ DNA quality; abnormal protamine 1:2 ratio (atypical protamine 2 → DNA packaging defect); additive toxicity with varicocele	Earlier conflicting; recent data argue impairment
Maternal smoking	Aryl hydrocarbon receptor ligand → fetal testis apoptosis (blocked by antagonist); sons have smaller testes, lower counts, altered hormones ; altered secondary sex ratio (possibly via maternal T)	Laboratory and epidemiological concordance
Electronic cigarettes	Lower carcinogen/toxicant levels than smoking; no conclusive data on androgens or semen	No long-term data

Concept — most spermatotoxic drugs converge on one pathway — **raising oestrogen or supplying androgen suppresses pituitary LH** and hence intratesticular testosterone — which is why exogenous testosterone, the most "androgenic" of therapies, is itself a contraceptive.

Infections and Inflammation

- Infection of testis, epididymis, prostate and urethra impairs fertility by **anatomical and functional** means.

COMPARISON — ORGANISMS BY REPRODUCTIVE SITE

Site	Typical organisms
Prostate	<i>Escherichia coli</i> , <i>Pseudomonas aeruginosa</i> , <i>Klebsiella</i> , <i>Proteus</i> , <i>Enterococcus</i>
Epididymis	<i>Neisseria gonorrhoeae</i> , <i>Chlamydia trachomatis</i> , <i>E. coli</i>
Urethra	<i>N. gonorrhoeae</i> , <i>C. trachomatis</i> , <i>Mycoplasma</i> , <i>Trichomonas vaginalis</i>
Testis (infrequent)	Mumps (<i>Rubulavirus</i>), coxsackievirus B, <i>N. gonorrhoeae</i> , <i>C. trachomatis</i> , <i>E. coli</i> , <i>P. aeruginosa</i> , <i>Klebsiella</i> , <i>Staphylococcus</i> , <i>Streptococcus</i>
Any organ	<i>Mycobacterium tuberculosis</i> — scars vas and epididymis; infrequent in industrialised nations

- **Mumps orchitis** is so painful and bizarre that even a young boy is unlikely to forget it — the history is reliable.
- Consequences pair: **anatomical** (e.g. urethritis → stricture) versus **functional** (direct sperm damage, or immune induction in any reproductive organ).
- Direct effect: **C. trachomatis serovar E** elementary bodies degrade sperm DNA in a time-dependent manner. *E. coli* harms sperm in vitro, but **most bacteria, including *E. coli*, have little in vivo effect on motility**; sexually transmitted organisms may be more virulent — differential effects remain unclear.
- **Viruses** may integrate into the genome and pass vertically through the germline. **Hepatitis C and human immunodeficiency virus** nucleic material appears in seminal plasma but is **not correlated with impaired sperm function**; **human papillomavirus** is associated with impaired bulk parameters, and in vitro heparinase III lowered viral load without harming sperm function.
- Indirect mediators: leucocytosis, reactive oxygen species (ROS), interleukin-1 (IL-1), IL-6, IL-8, interferon- γ , macrophage migration inhibitory factor, tumour necrosis factor- α , epididymal macrophages, dendritic cells — the immune system may lose self-recognition of sperm or overwhelm sperm defences.
- **Non-infectious or post-infectious prostatitis** may impair sperm via **pyospermia and ROS, antisperm antibodies**, or altered prostatic ions (zinc, magnesium, calcium, selenium); prostatitis can generate ROS **without leucocytosis**. How much inflammation adds beyond infection is unknown.

Concept — infection damages fertility along two routes, **anatomical scarring and functional sperm injury**, and in the functional route the host's inflammatory and oxidative response matters as much as the organism itself.

Childhood Diseases

- Early maladies either **misroute or obstruct** the gamete's path from testis to female tract, or **disturb production** or damage mature sperm.

Pediatric Surgery

- **Childhood hydrocele and hernia repair**: low but discrete incidence of **vasal obstruction**; testis atrophy after paediatric inguinal herniorrhaphy is rare.
- **Adolescent hernia repair often uses mesh** — consider mesh-related vasal occlusion in an infertile man with this history.
- **Posterior urethral valve ablation**: scarring and dysfunction in earlier series; rarely seen now.
- **Older bladder neck reconstructions** caused retrograde ejaculation; rarely performed today.

Testis Torsion

- In males ≤ 25 years, torsion is **more than three times commoner than testis cancer**.
- Contralateral biopsy is **abnormal in most** — implying either **unnoticed prior torsion** or an **underlying pathology** expressed both as abnormal scrotal anatomy and as spermatogenic dysfunction.
- About **half** develop adverse spermatogenic effects; a minority end with low sperm concentration.

- Torsion disrupts Sertoli cell tight junctions (the **blood–testis barrier**), so some develop **antisperm antibodies**.

Cryptorchidism

Descent sequence: gonadal precursor cells arise on the posterior abdominal wall in **gestational week 5** → differentiation of testicular cell types and compartments, development of the testicular envelope and its connection to distal ducts → migration towards the nascent inguinal canal → scrotum; completed only in the **third trimester**.

- Regulators (animal models): **insulin-like 3 (INSL3)**, its receptor **relaxin/insulin-like family peptide receptor 2 (RXFP2)**, **antimüllerian hormone (AMH)**, **HOXA10** and other HOX genes.
- Some defects only **arrest mechanical descent**; others also impair **spermatogenesis** — so infertility exceeds the thermal toxicity suffered by an undescended testis.
- **Androgens induce regression of the cranial suspensory ligament** in the fourth gestational month, permitting descent.
- Undescended testes affect **up to 4% of term newborns**, falling to **below 1.5% by 1 year**. Twin and sibling concordance suggests **maternal inheritance plus intrauterine environment**. **Unilateral is about twice as common as bilateral**.

COMPARISON — UNILATERAL VERSUS BILATERAL CRYPTORCHIDISM (AFTER ORCHIDOPEXY)

	Unilateral	Bilateral
Paternity	Near that of controls , regardless of age at surgery or testis size	Markedly reduced
Inhibin B (Sertoli marker)	About two-thirds of controls	About one-third of controls
Sperm density	Lower than normal	Lowest
Ultrastructural sperm defects	More than controls	Most

- **Laterality is the key prognostic distinction**. T differences are smaller than inhibin B differences → the lesion lies in the **seminiferous epithelium more than in Leydig steroidogenesis**.
- **Timing:** post-pubertal orchidopexy barely changes bulk semen parameters. AUA guideline (2014): **repair within the first 18 months**.
- About a **quarter** of cryptorchid boys are born with **reduced germ cells**; after **15–18 months** some biopsies lack germ cells, rising to a large fraction by 8–11 years.
- Serum T falls with **later age at orchidopexy** → earlier surgery spares Leydig function.
- The old view that germ cells stay quiescent until puberty (so any prepubertal age is equivalent) is wrong: the endocrine axis matures and the **fetal-to-adult spermatogonial transition occurs very early**.
- Surgical sperm retrieval in azoospermic men with prior orchidopexy: studies conflict, but one series found **far better retrieval with orchidopexy before about 10 years**, and ROC analysis ranks **age at orchidopexy second only to T** in predicting retrieval.
- Recommendation: **before 18 months**; if missed, **before 10 years**. Men beyond that threshold **may still have sperm retrieved** for ICSI.

- **Ascending testes** (descended, later ascend — occasionally to intra-abdominal): fertility prognosis **unknown**. **Retractile testes**: sperm present but density lower than expected, **approaching that of cryptorchidism**.

Concept — cryptorchidism is fundamentally a **germ-cell and Sertoli-cell lesion** that begins in fetal life and worsens with time, so the number of affected testes predicts fertility and the age at repair determines how much germinal and Leydig reserve is salvaged.

Genetics

- Ask whether **blood relatives had difficulty conceiving**, and about family syndromes linked to reproductive dysfunction such as **cystic fibrosis (CF)**.

Sexual History

- Lifestyle may reduce intercourse frequency below that needed; optimum is **daily around ovulation**.
- **Mittelschmerz** allows some women to predict ovulation, but many misread bodily sensations — **symptoms are unreliable**.

Timing method: basal body temperature or home hormonal kits detect ovulation **only after it occurs** → record the day of ovulation for **2–3 cycles** → begin daily intercourse **several days before the earliest recorded day**. Impractical with **advanced maternal age** (delays therapy) → more aggressive strategies with the female fertility specialist.

COMPARISON — LUBRICANTS AND SPERM

Impair sperm	Sperm-safe
K-Y Jelly, Keri Lotion, Astroglide, FemGlide, Replens; saliva (toxic). FemGlide, Replens, Astroglide ↓ motility; FemGlide and K-Y Jelly ↓ chromatin quality	Pre-Seed (isotonic): no significant loss of motility or chromatin integrity (acridine orange sperm chromatin structure assay); also safe during semen collection — no effect on bulk parameters, membrane integrity, ROS, antioxidant capacity or DNA integrity

- Ask about **erectile function**: without intercourse, semen is not deposited near the cervical os (Chapters 71–72).
- Infertility carries a heavy **psychological burden**. Intercourse frequency is altered in **up to half** of males in treatment and tracks libido and satisfaction — a revealing question is **whether coital frequency has changed** during evaluation.
- Coping differs by sex: **males distance themselves and problem-solve**; **females seek social support** — each may misread the other's normal strategy as a problem.
- That males **conflate fertility with masculinity** is a common misconception; it happens only infrequently.
- **Stress may impair semen quality** — a vicious circle; once in therapy (including IVF), a male-factor diagnosis **does not disturb well-being** or well-adjusted relationships.
- Consider referral to a psychologist; framed as **problem-solving**, many men engage readily.

KEY POINTS: MALE REPRODUCTIVE HISTORY

- The most important determinant of a couple's reproductive potential is **maternal age**.
- Many conditions affect male reproductive function; organise the history into **toxicants, infectious processes, childhood conditions and sexual history**.

Concept — because reproduction needs two patients and is probabilistic, the history modifies a **probability of conception** rather than establishing a diagnosis — **timing, frequency and female age set the baseline, and the TICS exposures adjust it**.

PHYSICAL EXAMINATION

General Physical Examination

- Male and female faces differ; **female facial features** suggest sex chromosomal and androgenisation disorders.
- Reduced **facial, truncal, axillary and pubic hair** suggests inadequate androgenisation; pubertal androgen failure → **high-pitched voice**; excess endogenous or therapeutic E2 → **gynaecomastia**.
- **Eunuchoid habitus** (epiphyses fail to close from low pubertal T): **arm span >5 cm longer than height**; **lower segment (pubis–heel) >5 cm longer than upper segment (crown–pubis)**.
- **Klinefelter syndrome** is classically described with gynaecomastia, eunuchoid appearance and tall stature — but **many 47,XXY males lack this habitus**.

Obesity

- **↑ E2** from peripheral aromatisation of T in abundant adipose tissue. A **TTTA aromatase polymorphism** links rising E2 to rising body mass, and carriers show the greatest fall in E2 with weight loss.
- **↓ T**, with four proposed causes: **E2 negative feedback → ↓ LH; ↑ leptin; insulin resistance; sleep apnoea**. LH data are inconsistent → the T deficit may be **gonadotropin-independent**.
- **↓ SHBG** (attributed to hyperinsulinaemia) → **bioavailable T exceeds what total T predicts**; an obese man may be more androgenised than his total T suggests.
- Serum **inhibin B falls with body mass index (BMI) in adult but not prepubertal males** → obesity reduces Sertoli cell number **during puberty**.
- **Endocrine-independent effects**: **↓ paternity with ↑ BMI independent of hormones**; only extreme obesity may act endocrinologically; some studies link BMI to motility but not concentration (implicating **epididymal function**); obesity may degrade **sperm DNA integrity and mitochondrial activity**.

Concept — *obesity impairs fertility on two tracks at once, an **aromatase-driven endocrine track** and a **hormone-independent track** acting on the epididymis and sperm DNA, so correcting hormones alone need not restore function.*

Male Reproductive Physical Examination

- Most of the system is external and palpable. Examine **standing**, before a low table or chair — some men **faint during scrotal palpation**; asking about his work distracts him.
- The partner may add history, but the examination is a tactful moment to **ask her to leave** so the man can speak privately.

Examining the Scrotum

- **Hypoplastic hemiscrotum** → contents absent since birth.
- **Asymmetric enlargement** → reactive hydrocele or tumour.
- **Visible varicocele**.
- Scrotum lying against the thighs in a large or obese man → **insufficient scrotal–body temperature gradient**.

Examining the Testis and Epididymis

- The epididymis is **normally hard to feel**; an **easily palpable epididymis is engorged → obstruction**.
- Upper pole discernible but lower pole not → **incomplete wolffian duct development**.
- **Testis size correlates with sperm production**.

COMPARISON — METHODS OF MEASURING TESTIS SIZE

Method	Technique	Abnormal
Seager caliper orchidometer (Fig. 69.1)	Long axis gently grasped between the jaws	Long axis ≤ 4.6 cm → spermatogenic impairment
Prader orchidometer (Fig. 69.2)	Palpation compared against a string of ellipsoids of marked volume	Volume ≤ 20 mL is low
Scrotal ultrasonography	Direct measurement	More accurate; clinical value of the increment unclear

Examining the Spermatic Cord

- Two findings of reproductive significance: **is the vas present**, and **is there a varicocele**.
- The vas is **firm and cord-like**; vessels are **compressible**. Veins can mimic the vas, so absence is a difficult sign.
- **Meacham's maxim** (Randall Meacham): search for the vas as in the first step of vasectomy, delivering it to the skin surface; **if the presumed vas slips from the fingers three times, it is absent**.
- **Unilateral absence** → possible complete ipsilateral wolffian failure, **including renal agenesis** → consider **renal ultrasound**.
- **Bilateral absence** → **high likelihood of a CF gene mutation** → sequence the **cystic fibrosis transmembrane conductance regulator (CFTR)**; renal agenesis occurs in about **one in ten** with congenital bilateral absence of the vas deferens (CBAVD) → renal ultrasound too.
- Inspect the upper scrotum for a visible venous plexus, then gently palpate for a **varicocele**.
- History: sporadic pre-1955 reports; **W. Selby Tulloch (1955)** first systematically reported high ligation in infertile men with improved counts; **Dubin and Amelar** studied larger series and educated urologists on its pathology and treatment.
- Varicocele is the **commonest nonductal, surgically correctable** lesion affecting male fertility: about **1 in 5–6 of the general population** versus **1 in 3 to 1 in 2 of infertile men**. That **not all men with a varicocele are infertile** is one of the field's most perplexing problems.
- Many 3–4-grade systems exist; a sensible one separates **subclinical** (treatment does not help) and **seen from only felt**.

COMPARISON — MODERN EVIDENCE-BASED VARICOCELE GRADING

Grade	Definition	Treat?
Subclinical	Not palpable or visible; detected only radiographically (e.g. Doppler ultrasound)	No
Grade I	Palpable only standing with Valsalva	Clinical
Grade II	Palpable but not visible	Clinical
Grade III	Visible through the scrotal rugae	Clinical

Examining the Phallus

- Semen must be deposited **near the cervical os**; note **phimosis**, **meatal displacement** (hypospadias, epispadias) and **significant curvature**.

Examining the Prostate and Seminal Vesicles

- Digital rectal examination (DRE) **adds little**; it may be **omitted if the patient is apprehensive**.
- If done: an **aplastic or hypoplastic prostate** suggests congenital malformation or significant hypoandrogenism.
- Seminal vesicles are **normally impalpable**; **palpable vesicles are abnormal** → engorgement and possible **ejaculatory duct obstruction (EDO)**.

KEY POINTS: MALE REPRODUCTIVE PHYSICAL EXAMINATION

- Obesity impairs male reproductive potential by **endocrine-dependent and endocrine-independent** mechanisms.
- **Testis size directly reflects spermatogenic mass.**
- **Unilateral** vasal absence suggests a wolffian duct anomaly; **bilateral** absence is associated with **CF gene mutations**. In both, renal agenesis may result.

Concept — the examination reads three things directly: **androgenisation** (habitus, hair, breasts), **spermatogenic mass** (testis size) and **ductal integrity** (epididymis, vas, seminal vesicles) — each points to a different compartment of the reproductive tract.

LABORATORY EVALUATION OF MALE INFERTILITY

- Three assessments: **endocrine evaluation**, **semen analysis** (the only direct appraisal of the gametes) and **genomic assessment** (a growing field).

Endocrine Evaluation

- Spermatogenesis depends on **intratesticular T**, so hypoandrogenism is associated with infertility. Adequate androgenisation threshold: **280 or 300 ng/dL** total T.
- Low T is common in men with **spermatogenic azoospermia**, in **oligospermia**, and even in a substantial share of clinic men with density **above 20 million/mL** (the 4th-edition WHO threshold).
- Most men with density ≤ 22 million/mL do not conceive within a year, so many below it have pathology — in about **one-third, endocrinopathy**. Therefore **assess androgenisation in every man presenting with infertility, including those above 20 million/mL**.
- No upper limit exists above which endocrine dysfunction is excluded; the WHO 50th percentile (**66 million/mL**, 2021 analysis) is a reasonable guide above which a full endocrine evaluation is unnecessary.

COMPARISON — CIRCULATING FORMS OF TESTOSTERONE (HEALTHY MAN)

Form	Binding	Share of total T	Bioavailable?
SHBG-bound	Tight	30–44%	No
Albumin-bound	Loose	54–68%	Yes
Free	Unbound	0.5–3.0%	Yes

- **Bioavailable T = free + albumin-bound**. With T 300 ng/dL and a lower bioavailable fraction of 54.5%, the lower limit of bioavailable T is about **164 ng/dL**.
- SHBG varies with disease and age (Box 69.1) → **total T cannot gauge bioavailable T**. Free T assays are difficult → **calculate bioavailable T from total T, SHBG and albumin** (Vermeulen formula); the International Society for the Study of the Aging Male provides an online calculator.

BOX 69.1 CONDITIONS ASSOCIATED WITH ALTERED SEX HORMONE-BINDING GLOBULIN (SHBG) CONCENTRATIONS

Decreased SHBG	Increased SHBG
Obesity	Ageing
Nephrotic syndrome	Hepatic cirrhosis and hepatitis
Hypothyroidism	Hyperthyroidism
Glucocorticoids, progestins, androgenic steroids	Anticonvulsant therapy
Acromegaly	Oestrogen therapy
Diabetes mellitus	Human immunodeficiency virus disease

Modified from Bhasin et al., Endocrine Society clinical practice guideline, 2010.

- **Pattern:** the "↓" column is dominated by **insulin-resistant and androgen/glucocorticoid-exposed states**; the "↑" column by **ageing, liver disease and oestrogenic states**.
- **Trap:** in obesity total T under-reads androgen status; in ageing, cirrhosis or HIV total T over-reads it.

COMPARISON — CIRCADIAN RHYTHMS

Measure	Young healthy men	Older men
Total T	Peak early morning , trough late afternoon	Level and rhythm attenuated
SHBG	Opposite: peak late afternoon, trough early morning (all ages)	Same opposing rhythm
Bioavailable T	Marked rhythm, early-morning peak	Rhythm and level substantially diminished

- Sample **in the morning** in all men to standardise; timing matters **most in younger men**.
- **LH localises hypoandrogenism:** ↑ LH → testicular (Leydig) failure; ↓ LH → pituitary. LH may be drawn after a low T or simultaneously.
- T and LH are **pulsatile**: investigate borderline values with **3 morning samples at 20-minute intervals**, assayed separately and averaged (historically pooled).
- Sertoli cells secrete **inhibin B (inhibits)** and **activin (stimulates)** follicle-stimulating hormone (FSH). Germ-cell loss → ↓ inhibin → ↑ FSH; FSH is thus an **indirect index of germ-cell mass**.

COMPARISON — AZOOSPERMIA: OBSTRUCTION VERSUS SPERMATOGENIC DYSFUNCTION (FSH AND CALIPER CRITERIA)

	Obstructive azoospermia	Spermatogenic dysfunction
FSH	≤7.6 IU/L	>7.6 IU/L
Testis long axis (caliper)	>4.6 cm	≤4.6 cm
Accuracy	Combined FSH and caliper long axis accurately predict the cause of azoospermia	

- When sperm **are present**, the odds of abnormal concentration rise sharply at **FSH 4.5 IU/L** — a second, lower threshold.

COMPARISON — SERTOLI-CELL MARKERS OF SPERMATOGENESIS

Marker	Nature	Status
FSH	Pituitary; indirect	Standard, clinically useful
Inhibin B	Sertoli product; direct	Often slightly more accurate , especially at low ranges; increment small → choose FSH or inhibin B by cost and availability
AMH	Sertoli product; transforming growth factor-β family	Encouraging pilot data, small sets → experimental

- **Aromatase** (testis, adipose tissue, liver, brain) converts T to oestrogens. Elevated E2 is proposed to impair fertility; a **T (ng/dL) to E2 (pg/mL) ratio <10:1** suggests dysfunction.
- E2 cuts both ways: **low E2 may also impair fertility**; E2 is inhibitory in somatic cells but **stimulatory in germ cells** (dose-dependent, non-additive proliferation), and **absent aromatase** causes dysfunctional spermatogenesis.
- **Prolactin** inhibits gonadotropins and T; raised by pituitary hyperplasia, adenoma or tumour. Assay it when **visual field change, headache or erectile dysfunction** accompany infertility, especially with low T. Clinically significant prolactinoma is **very rare** in infertile men → **not a routine initial test**. Prolactin is **labile** — repeat if raised.

- Thyroid-stimulating hormone, adrenocorticotrophic hormone or growth hormone: only if a **space-occupying pituitary lesion** is suspected; thyroid, cortisol or growth hormone assays if signs appear (exophthalmos, striae, moon facies, facial bony change) — not in initial screening.
- **Congenital adrenal hyperplasia (CAH): testicular adrenal rest tumours** and later infertility; monitor therapy with 17-hydroxyprogesterone, $\Delta 4$ -androsteredione, renin and T.

Initial endocrine screen (morning): total T + SHBG + albumin (calculate bioavailable T) → LH and FSH (pituitary) → E2 (aromatisation).

Concept — *one morning sample reads the whole axis: **bioavailable T** says whether there is a deficit, LH says where it arises, FSH (or inhibin B) says how much germ-cell mass remains, and E2 says whether aromatisation is distorting the loop.*

Evaluation of Semen

- Reproduction is probabilistic: the more viable sperm, the greater the chance of fertilisation. The **only definitive result is azoospermia** — only then is a man absolutely sterile.
- **MacLeod (1951)**: first rigorous statistical comparison of semen from fertile and infertile men (cumulative probability histograms, quartiles for concentration, motility, shape). The distributions **largely overlap**. Low values suggest infertility, but **values above a threshold do not confirm fertility**.
- The field wrongly assumes a man above threshold (e.g. >15 million/mL) is fertile; **the converse does not hold**.
- Remedy for overlapping distributions: **two thresholds** — below the lower, subfertility is probable; above the upper, fertility is probable; between, no prediction. **Guzick (2001)** used classification and regression tree (CART) analysis on fertile men versus men whose wives underwent intrauterine insemination (IUI) with female factor largely excluded.

Bulk Semen Parameters and the World Health Organization Criteria

- WHO laboratory manual editions 1–4: thresholds by expert panel and survey (e.g. **20 million/mL**) — fertile men fall below and infertile men above.
- **5th and 6th editions**: percentile tables of men whose partners conceived **within 1 year** of stopping contraception (Table 69.2). Limitations: derived from a **fertile, not infertile, population**, and **descriptive statistics cannot predict** outcome.
- The 6th edition also lists **5th percentiles with 95% confidence intervals (CIs)**, which has spurred their **misuse as new thresholds**. Best use: show the patient his values against a fertile reference. Pragmatically, communicate the **5th percentile as likely infertility** and the **50th as typical** of conception within a year.

TABLE 69.2 BULK SEMEN ANALYSIS PARAMETER PERCENTILES

Parameter	2.5th	5th (95% CI)	50th
Semen volume (mL)	1.0	1.4 (1.3–1.5)	3.0
Sperm concentration (million/mL)	11	16 (15–18)	66
Total number (million/ejaculate)	29	39 (35–40)	210
Total motility (%)	35	42 (40–43)	64
Progressive motility (%)	24	30 (29–31)	55
Normal forms (%)	3	4 (3.9–4.0)	14
Vitality (%)	45	54 (50–56)	78

Modified from Campbell et al., Distribution of semen examination results 2020, Andrology 2021 (WHO 6th edition data).

- **What the table is for**: locating a patient against men who conceived within a year — a reference, not a diagnostic cut-off.

- **Trap:** treating the 5th percentile as a pass/fail line; a value above it does not confirm fertility.

COMPARISON — SEMEN REFERENCE SYSTEMS

Parameter	Legacy WHO (1st–4th)	WHO 6th: 5th / 50th percentile	Guzick CART: subfertile below / fertile above	Guzick ROC AUC
Concentration (million/mL)	20	16 / 66	13.5 / 48.0	0.60
Motility (%)	—	30 / 55 (progressive)	32 / 63	0.59
Strict morphology (%)	—	4 / 14	9 / 12	0.66
Volume (mL)	—	1.4 / 3.0 (practical threshold 1.0)	—	—
Total count (million)	—	39 / 210	—	—

- **Pattern:** every parameter has an AUC near 0.6 — **poor discrimination**; morphology is marginally the best of three weak discriminators.
- Semen is **highly variable**: obtain **at least 2 analyses, 2–3 weeks apart**, with **constant abstinence**, since parameters decline as abstinence lengthens.

COMPARISON — ABSTINENCE INSTRUCTIONS

Historical (WHO)	Current evidence
2–7 days after the last ejaculation	1–2 days — parameters peak, then decline

- Collect into a **non-toxic wide-mouthed glass or plastic cup**; if religion or culture forbids masturbation, use a **special non-toxic condom**.

Macroscopic sequence: semen forms a coagulum → **liquefy 30 minutes** → viscosity by pipette aspiration (**drop thread ≤ 2 cm**) → colour inspection → microscopy in a defined-volume chamber (haemocytometer or Makler chamber).

SEMEN COLOUR

Colour	Meaning
White or light grey	Normal
Yellow or green	Infection, jaundice, vitamins or medication
Brown	Often spinal cord injury
Red	Blood

- **Semen pH is no longer recommended:** altered by environment, and its original use to detect obstruction is undermined by the vast size difference between a hydrogen ion and a sperm head.

Semen Volume

- Often unreported by low-volume laboratories, yet **clinically important**.

CAUSES OF SEMINAL HYPOVOLAEMIA (AND ASPERMIA)

Category	Causes
Anatomical	EDO ; prostatic and seminal vesicle hypoplasia (severe androgen deficiency, CBAVD)
Functional	Retrograde ejaculation
Neurological	SCI, DM, MS
Pharmacological	α-adrenergic blockers (e.g. tamsulosin)

- Practical threshold to **start evaluating hypovolaemia: 1.0 mL** (the 2.5th percentile).

- **Aspermia** (dry ejaculate, anejaculation): no fluid at orgasm; same causes.

Aspermia or hypovolaemia work-up: void → ejaculate for semen analysis → void again into a separate container → centrifuge urine and count sperm in the pellet (**postejaculatory urinalysis**) → plus transrectal ultrasonography (TRUS) or equivalent for **EDO**.

- Few urinary sperm with a large antegrade count are of **little consequence**; retrograde ejaculation is **clinically significant when urinary sperm near or exceed antegrade sperm**.
- **Hypervolaemia (>5 mL)** is rare; thought to **dilute sperm** → process into a smaller volume and perform **IUI**.

Sperm Density

- Reported in million/mL. **Oligospermia (oligozoospermia)** = low density; **cryptozoospermia** = too few to measure reliably.
- The legacy 20 million/mL threshold was often taken to define oligospermia.
- Oligospermia is now defined **in clinical context**: a man with one sample of 10 million/mL who impregnates his wife easily may not be oligospermic, whereas one with **small testes, raised FSH and repeated 20–25 million/mL** reasonably is.
- **Total sperm count = volume × density (millions)**.

Sperm Motility

- Assess **within 30 minutes of liquefaction**; low motility = **asthenospermia (asthenozoospermia)**.
- WHO 6th categories: **progressive** ("moving actively, either linearly or in a large circle, regardless of speed"), **non-progressive** (all other movement without progression), **immotile**.

Sperm Morphology

- Human sperm are **highly pleomorphic** — bizarre forms outnumber well-formed ones in any man. Excess abnormal forms = **teratospermia (teratozoospermia)**.
- Earlier WHO criteria were generous yet still classed most sperm abnormal. **Kruger (strict, Tygerberg) morphology**: several features assessed, and **any one out of range makes the sperm abnormal** → thresholds around **5%** characterise a normal ejaculate. The **WHO 6th edition adopted strict morphology**.
- **Predictive value is questionable**: possibly linked to embryo formation, but most studies show strict morphology is **unrelated to sperm nuclear integrity** and **does not predict natural conception, IUI or IVF outcome**.
- Technicians scrutinising more closely report ever-lower normal percentages → many men now labelled with **isolated teratozoospermia likely have adequate fertility**.
- Specific monomorphic defects clearly matter: **globozoospermia** (acrosome fails to form → small round heads); **pinhead sperm** (basal plate fails to attach to the nucleus opposite the acrosome during spermiation → heads absorbed, tails only).

Sperm Vitality

- Vitality = proportion of **metabolically active living** sperm; many dead sperm = **necrospermia (necrozoospermia)**.
- Essential with **near-total or total asthenospermia**, to distinguish **cell death from a defect of motility machinery**.

COMPARISON — VITALITY TESTS

	Eosin Y ± nigrosin	Hypo-osmotic swelling (HOS) test
Principle	Live sperm exclude eosin ; dead sperm absorb it; nigrosin darkens the background for contrast	Live sperm with intact membranes show tail swelling within 5 minutes in hypo-osmotic medium

	Eosin Y $\pm$ nigrosin	Hypo-osmotic swelling (HOS) test
Sperm afterwards	Destroyed — diagnostic only	Preserved — usable for IVF/ICSI

Concept — the semen analysis is a **probability estimate, not a verdict**: fertile and infertile distributions overlap, so a low value suggests infertility, a normal value never proves fertility, and only azoospermia is definitive.

Genomic Assessment

- Irony: genes passed from father to son may cause a condition that, untreated, would stop them being passed on; genetics plays a significant role in male reproductive dysfunction.

Karyotype

- Methods: classical **dye banding**; **fluorescence in situ hybridisation (FISH)** with sequence-specific probes; **spectral karyotyping (SKY)** colouring every chromosome.
- ASRM/AUA guideline (2021): genetic testing including **karyotype in all men with azoospermia from spermatogenic dysfunction and in severe oligospermia (<5 million/mL)**.
- Anomaly rates vary by region and cost may be significant → the physician may judge its indication in his or her population.

Y Chromosome Microdeletion Testing

- The Y is among the smallest chromosomes (**about 60 megabase pairs**), determines male sex, and is the **only chromosome passed directly father to son**; it has a male-specific region and two pseudoautosomal regions (**PAR1, PAR2**).
- **Tiepolo and Zuffardi (1976)** located a region on the **long arm (Yq)** critical for spermatogenesis — the **azoospermia factor (AZF)**.
- About **95% of the Y does not recombine**; about a third of that is **amplicons** (palindromic repeats) — a substitute repair mechanism that also creates **fragility** → **microdeletions**.

COMPARISON — AZOOSPERMIA FACTOR REGIONS

Region	Position	Genes	Surgical sperm retrieval
AZFa	Distant and isolated	USP9Y, DDX3Y	Complete deletion → essentially nil
AZFb	Overlaps AZFc	EIF1AY, RPS4Y2, KDM5D	Complete deletion → essentially nil
AZFc	Overlaps AZFb; proximal part sometimes called AZFd (utility unclear)	DAZ genes	Impairment, not failure ; sperm often in ejaculate or testis
gr/gr (AZFc subregion)	Within AZFc	—	Clinical relevance unclear

- Reasonable to **test azoospermic men before surgical sperm extraction** to counsel on retrieval odds — equally reasonable to omit, given the **rarity of AZFa and AZFb deletions**.

Cystic Fibrosis Transmembrane Conductance Regulator Mutation Assessment

- CFTR encodes a **chloride (and possibly bicarbonate) channel** that may regulate other ions.
- **More than 1600 mutations**, classed **mild or severe** by whether full CF results. Commonest severe: **Δ F508** — a 3-base-pair deletion removing phenylalanine at position 508. Many patients harbour more than one mutation.
- **A severe mutation on each allele** → **a child with CF** → screening of **both prospective parents** is imperative when CFTR alteration is suspected.

- Screening panels cover **25–40 common mutations** → a **negative panel still carries residual risk**; full-gene testing exists but costs more.
- Prevalence varies by **ethnicity and geography** → interpret with location and ethnicity; reports are usually ethnicity-stratified.

KEY POINTS: LABORATORY EVALUATION OF MALE INFERTILITY

- Endocrine assessment includes **total T, the portion not bound to SHBG, E2, and LH and FSH**.
- The semen analysis is a **probabilistic** assessment; aside from azoospermia, **no threshold absolutely separates fertility from infertility**.
- Semen volume **<1.0 mL**: differential includes **EDO, retrograde ejaculation, and vasal and accessory gland maldevelopment (CBAVD)**.
- Vitality staining distinguishes **complete asthenospermia from necrospermia**; common staining methods distinguish **pyospermia from immature germ cells**.
- **Globozoospermia** (round heads) indicates deficient acrosome formation; treatment is **IVF with ICSI**.
- Blood–testis barrier disruption causes **antisperm antibodies**, whose significance depends on the **degree of binding to sperm heads**.
- CFTR screening of men with CBAVD **and their partners** identifies severe mutations such as $\Delta F508$ that may cause overt CF in offspring.

Concept — genetic tests are chosen by phenotype — **karyotype and Y microdeletions for failed or severely impaired production, CFTR for absent or obstructed ducts** — and their yield is counselling on retrieval odds and offspring risk rather than treatment.

IMAGING IN THE EVALUATION OF MALE INFERTILITY

- Imaging is **infrequently needed and should be ordered cautiously**: incidental benign findings such as **testicular microlithiasis** cause distress and unnecessary testing. The description here is not an indication for routine screening.

Transrectal Imaging

- Consider EDO when **azoospermia accompanies low semen volume**.
- **TRUS** is the earliest and still most prevalent test. EDO signs: **seminal vesicle anteroposterior diameter >1.5 cm**, with or without a **midline prostatic cyst** (Fig. 69.5 — intraprostatic cyst, transverse and longitudinal views).
- TRUS has **low specificity**: EDO may exist without vesicle dilatation, and **fertile men may have dilated vesicles depending on abstinence**.

COMPARISON — INVESTIGATIONS FOR EJACULATORY DUCT OBSTRUCTION

Test	Method	Standing
TRUS	Vesicle diameter, midline cyst	Convenient, common; low specificity
Vesiculography	Contrast injected directly into the seminal vesicles	More accurate in a small series; invasive, costly
Seminal vesicle aspiration	Aspirate examined for sperm; ≥3 sperm per high-power field (400×) is positive	Useful for partial EDO
Chromotubation	Diluted indigo carmine or methylene blue into the vesicles; cystoscopic check for dye at the verumontanum ductal orifices	More accurate in a small series; invasive, costly

Test	Method	Standing
MRI; hydraulic pressure measurement	—	Additional options

- Any incremental accuracy of the invasive tests over TRUS, even if confirmed in larger studies, **may not justify the added risk and cost.**

KEY POINT: IMAGING

- Imaging can reveal sequelae of genetic conditions (**vasal absence, renal aplasia**) and distinguish causes of hypovolaemia such as **EDO**, but is **infrequently necessary** to diagnose varicocele or spermatogenic dysfunction.

Concept — *imaging answers one narrow question — **is the duct system obstructed or absent** — and is not a screening tool, because its incidental findings do more harm than its yield.*

ASSISTED REPRODUCTION

Methods for Sperm Selection

- ICSI needs only as many sperm as oocytes, and **bypasses the natural barriers** that normally select the fittest sperm.
- Established: **density gradient centrifugation** and **swim-up** — yield fast-swimming sperm with better morphology.
- Newer in vitro selection improves sperm quality but has **no conclusive evidence of better reproductive outcomes**; chemoattractants, microfluidics and optical systems are under investigation.

DIAGNOSES AND THERAPIES

- Pathophysiology remains incompletely understood and assessment is probabilistic, yet enough is known to make **reasoned judgements** about whether a male cause exists, its basis and plausible therapy.

Genetic Syndromes

- The human genome was completed in **2004**; broad carrier panels exist and whole-genome sequencing is used in research, but their use as **general screening tools is limited** by poor understanding of how spermatogenic genes act in concert.

Chromosomal Numerical Disorders

- **Klinefelter syndrome (47,XXY)** is the **commonest identified genetic cause** of male infertility; **1 in 500–1000 live births**.
- Adults: **azoospermia in the great majority, small testes, raised gonadotropins**; learning disability in most children; low T in most men.
- **Tall stature in only about 30%** → Klinefelter **cannot be excluded by examination**.
- ↑ **mediastinal germ cell tumours** → broader testicular effects.
- Karyotype: **non-mosaic 47,XXY in 80–90%**; remainder **mosaic 46,XY/47,XXY** or extra/structurally abnormal X.
- About **8% have ejaculated sperm**; about **50% have testicular sperm** retrievable for IVF/ICSI; **earlier diagnosis** carries a better prognosis.
- Management trap: boys found after puberty with low T are often given **exogenous T alone, suppressing residual spermatogenesis**. Because spermatogenesis declines progressively, some advocate **surgical sperm extraction in early to mid puberty, before T and aromatase inhibitor therapy**.

Structural Chromosomal Anomalies

- Y microdeletions: see the AZF comparison — **AZFa and AZFb** leave no retrievable sperm; **AZFc** is of unclear significance.
- Deletion carriers with **oligozoospermia** risk a **progressive fall in concentration** → consider **sperm cryopreservation**.
- **Ring Y chromosome, r(Y)**: terminal breaks in both arms with fusion; phenotype varies with material lost. **Autosomal karyotypic anomalies** may also cause infertility.

Testicular Causes

- Two compartments: **seminiferous tubules** (gametes) and **interstitium** (Leydig cells); both are needed, and sperm must then exit. Hence three testicular causes: **spermatogenic dysfunction, steroidogenic dysfunction, microductal obstruction**.

Spermatogenic Dysfunction

COMPARISON — HISTOLOGICAL PATTERNS OF SPERMATOGENIC DYSFUNCTION

Pattern	Meaning
Hypospermatogenesis	All stages present, production reduced
Maturation arrest	Sequence halts at a premeiotic, meiotic or postmeiotic step
Sertoli cell-only syndrome	Complete loss of spermatogonia

- **Histopathology** is the present means of assessment; genomic, proteomic and metabolomic markers are anticipated.
- In azoospermia, the **FSH and caliper criteria** (Laboratory Evaluation) identify spermatogenic dysfunction.

Antisperm antibodies

- Sertoli **tight junctions form the blood-testis barrier**, protecting haploid germ cells; disruption exposes spermatids and spermatozoa to antibody formation → **agglutination, impaired motility, reduced fertilising potential**.
- Causes: **obstruction** (e.g. after vasectomy); **inflammation** (orchitis, prostatitis, sexually transmitted infection); **heat** (varicocele, cryptorchidism, hot tubs); **trauma and torsion**; **genetic** (thymic maldevelopment, HLA-B28 haplotype).
- Treatment: **condoms and sperm washing lack good evidence** — washing removes unbound antibody, but the relevant antibody is bound. **Corticosteroids**: two controlled trials conflict. **IVF and ICSI** have become the common treatment.

Steroidogenic Dysfunction

- **Hypergonadotropic (primary) hypogonadism** = impaired T synthesis from Leydig cell dysfunction: **↑ LH, ↓ T**.
- Leydig failure may **coexist with pituitary insufficiency** → low T with variable, non-elevated LH; **ageing** combines falling androgen with a blunted pituitary response.
- **Intratesticular steroidogenesis is an absolute requirement**, especially for postmeiotic maturation.
- Klinefelter men often have low T, but Leydig failure may not be the only mechanism. Genetic causes: **LH receptor mutations** (humans), FSH receptor deficiency (mice); others — **orchitis, cytotoxic chemotherapy, environmental toxicants**.
- **No accepted therapy. Exogenous T is not indicated** — it cannot achieve intratesticular levels and suppresses LH.
- **Azoospermia + low T + markedly raised LH, desiring paternity** → **surgical sperm extraction**.

Microductal Obstruction

- Congenital or acquired obstruction of the **epididymis or scrotal vas**; **bilateral** → **azoospermia**.
- **Unilateral** obstruction: parameters may or may not fall, depending on the other testis; if the obstructed side has good spermatogenesis and the contralateral testis is impaired, parameters fall and **microductal reconstruction may be indicated**.
- **Herniorrhaphy, especially with mesh**, can obstruct the inguinal vas; bilateral occlusion → azoospermia. Reconstruction: Chapter 70.

Fig. 69.11 Algorithm for evaluation of azoospermia

Azoospermia → bilateral absence of vasa? **Yes** → **CFTR panel for patient and female partner**.

No → semen volume <1 mL? **Yes** → sperm absent on postejaculatory urinalysis? **No** (sperm present) → **retrograde ejaculation**; **Yes** → **TRUS for EDO**.

Volume normal → long axis >4.6 cm and FSH ≤7.6 IU/L → **obstruction likely**; long axis ≤4.6 cm and FSH >7.6 IU/L → **spermatogenic dysfunction likely**.

Concept — the testis fails in one of three places — **germ cells, Leydig cells or the outflow ducts** — and FSH, LH and testis size together localise the failure, which then dictates whether reconstruction, retrieval or neither is the answer.

Pituitary Dysfunction

- **LH drives Leydig steroidogenesis; FSH drives spermatogenesis via Sertoli cells**. Suppression of either, intrinsic or external, impairs spermatogenesis.

Hypogonadotropic Hypogonadism

- = decreased pituitary gonadotropin secretion. **Kallmann syndrome: anosmia with hypogonadotropic hypogonadism** (Kallmann and Schoenfeld, 1944); about **1 in 10,000 males**; inheritance most often **autosomal recessive**, also autosomal dominant and X-linked recessive.
- Genes: **KAL1** (anosmin-1, neurotropic factor in embryogenesis), **GNRHR** (gonadotropin-releasing hormone [GnRH] receptor), **PIT1**, **PROP1**, **GPR54** (kisspeptin receptor), homeobox **HESX1**, **LHX3**, **LHX4**; about 10% carry GNRHR or KAL1 mutations.

COMPARISON — HYPOGONADOTROPIC STATES

Condition	Hormones	Phenotype	Treatment
Kallmann / complete HH	↓ LH, ↓ FSH	Hypoandrogenism, azoospermia, ± anosmia	hCG ± rFSH or hMG
Isolated LH deficiency ("fertile eunuch")	↓ LH, normal FSH	Hypoandrogenised but produces sperm	Replace LH activity (hCG)
Isolated FSH deficiency	↓ FSH, normal LH	Androgenised but low sperm	Replace FSH
Isolated hypothalamic GnRH deficiency	↓ GnRH	Hypogonadotropic	Pulsatile subcutaneous GnRH pump every 2 hours, ≥6 months
Partial HH (common)	LH above Kallmann range but inappropriately low for the low T	Hypoandrogenism	SERMs or aromatase inhibitors

COMPARISON — DRUGS FOR HYPOGONADOTROPIC HYPOGONADISM

Agent	Class / role	Regimen	Notes
Human chorionic gonadotropin (hCG)	LH replacement	1500–5000 IU IM or SC 2–3 times weekly ; max 10,000 IU/week; titrate to serum T	May initiate spermatogenesis alone
Human menopausal gonadotropin (hMG)	FSH (with LH-like) activity	75 IU 2–3 times weekly , usually SC	Can be withdrawn months after spermatogenesis returns
Recombinant FSH (rFSH)	FSH replacement	—	As hMG
GnRH	Hypothalamic replacement	SC portable pump every 2 hours	Courses ≥ 6 months
Clomiphene citrate	Selective oestrogen receptor modulator (SERM)	25 mg daily or 50 mg alternate days ; titrate to T, max 100 mg daily	FDA-approved for women only; off-label in men >40 years; $\uparrow$ T without PSA or haematocrit change ; target normal T or 600–800 ng/dL
Tamoxifen	SERM	—	Pituitary stimulation
Anastrozole	Aromatase inhibitor	1 mg daily	Dosing not standardised
Letrozole	Aromatase inhibitor	—	As anastrozole

- Men diagnosed late, after **years of exogenous androgen**, may need **1–2 years of gonadotropins** before ejaculated sperm appear.

Prader-Willi syndrome

- Infantile failure to thrive with poor suck**, then **loss of satiety** in early childhood → marked obesity if uncontrolled; about **1 in 15,000–30,000**.
- Features: **hypogonadism**, **small testes**, dysmorphic facies, **growth hormone deficiency** (short stature, small hands and feet), pain insensitivity, obsessive-compulsive traits; hypogonadism and growth hormone deficiency presumed **hypothalamic**.
- Cause: **loss of paternally expressed genes at 15q11–q13** through failed **imprinting** (normally maternal copies silenced, paternal active; in Prader-Willi the paternal copies are also inactive).

Pituitary Tumors and Diseases

- Sellar lesions — secretory and non-secretory tumours, **craniopharyngioma** — compress the anterior pituitary → variable LH/FSH suppression.
- Prolactinoma** is the commonest relevant pituitary tumour, often with **erectile dysfunction**, but **rare** in infertile men → routine prolactin in asymptomatic men is questionable.
- Prolactin 20–50 µg/L**: no further evaluation; **significant elevation** → **cranial MRI**.
- Non-surgical prolactinoma: **dopamine agonists bromocriptine or cabergoline** — **cabergoline has fewer side effects**.

Other Pituitary Lesions

- Infiltration**: infective granulomas, **sarcoidosis**, **histiocytosis**; **iron deposition** (haemochromatosis, repeated transfusion).
- Systemic**: morbid obesity, chronic malnutrition, **type 2 DM**. Treat the underlying condition.

Extrapituitary Endocrine Modulators

- **Exogenous androgens (especially T)** suppress gonadotropins; **cannabis, antipsychotics, opioids and environmental toxicants** inhibit the pituitary via **oestrogenic and dopaminergic** pathways → **remove the agent**.
- **CAH**, especially milder forms presenting in adolescence or adulthood → hypogonadotropic hypogonadism plus frequent **testicular adrenal rest tumours**. **Corticosteroids may restore fertility**, but the long course needed for spermatogenesis brings weight gain and skin changes.

Concept — *gonadotropin deficiency is the most reliably treatable cause of male infertility because the testis itself is normal: **replace the missing upstream signal** — hCG for LH, FSH for FSH, pulsatile GnRH for hypothalamic failure, or unblock feedback with a SERM or aromatase inhibitor — and spermatogenesis follows.*

Extratesticular Endocrine Dysfunction

- **E2 inhibits gonadotropin release**; drugs raising E2 cause hypogonadotropic hypogonadism → **substitute another agent** where possible.
- **Obesity**: aromatase-rich adipose raises E2; other suspected links — the "neohormones" **leptin and kisspeptin, diabetes, sleep apnoea** — with or without hypogonadotropism.
- A large family cohort showed a **dose-response relationship between BMI and male infertility**, confirmed elsewhere.
- Obesity also **lowers LH pulse amplitude** and **blunts testicular response to LH**.
- Weight loss: rodent diet-and-exercise data positive; human data sparse and inconclusive — still **prudent to recommend** given certain general health benefit. Male obesity **worsens clinical pregnancy, miscarriage and live birth rates** in assisted reproduction.
- **Aromatase inhibitors** (anastrozole, letrozole, testolactone) raise T and lower E2; limited data suggest semen improves. Proposed indication: **T (ng/dL) : E2 (pg/mL) <10:1** = aromatase overactivity.
- **Caution with long-term aromatase inhibitors**: male **bone density may be E2-dependent**, and long-term data are lacking.

COMPARISON — ANDROGEN RECEPTOR DEFECTS VERSUS 5 α -REDUCTASE TYPE 2 DEFICIENCY

	Androgen receptor (AR) mutation	5 α -reductase type 2 deficiency
Locus	Xq11-12	2p23
Mechanism	Receptor insensitivity; longer CAG repeat in exon 1 associated with infertility	No conversion of T to dihydrotestosterone, which builds external male genitalia
Spectrum	Complete testicular feminisation → isolated male infertility	Female → male phenotype
Hormones	↑ T, ↑ E2, ↑ LH (variable), normal FSH	—
Clue	Markedly high T with impaired fertility → suspect AR resistance	Phenotypic females may harbour testes with intact spermatogenesis
Treatment	High-dose T (limited data); ICSI with ejaculated or testicular sperm	No medical treatment; ICSI pregnancies reported

Concept — *here the defect lies outside the gonad — **excess oestrogen silencing the pituitary, or a target tissue deaf to androgen** — so the laboratory signature (a low T:E2 ratio, or high T with high LH) points straight at the lesion.*

Developmental Disorders

- Maldevelopment of the **testes, external genitalia and microductal system** may present as adult infertility (disorders of sex development, hypospadias, epispadias: Chapters 43, 55, 56, 59, 60).

Cryptorchidism

- The **most significant prognostic feature is laterality** — unilateral or bilateral.

Microductal Aplasia

- The vas may fail on one or both sides; the **pathophysiology differs**.

COMPARISON — CONGENITAL UNILATERAL VERSUS BILATERAL ABSENCE OF THE VAS DEFERENS

	Unilateral (CUAVD)	Bilateral (CBAVD)
Basis	Ipsilateral wolffian (mesonephric) duct failure ; epididymis and ejaculatory duct may also be malformed	CFTR mutation spectrum (mild alleles), or unidentified mesonephric genes
Key association	Ipsilateral renal agenesis (renal and wolffian development coupled)	CF / CF carrier state ; renal agenesis in a minority
Work-up	Renal ultrasound	CFTR testing of man and partner ; renal ultrasound
Spermatogenesis	Depends on the other side	Typically normal
Fertility route	Contralateral system	Surgical sperm retrieval with ICSI

Congenital Bilateral Absence of the Vas Deferens

- Oates and colleagues (early 1990s): men with CBAVD have a **high frequency of CF-associated mutations**, and **nearly all men with CF lack both vasa** → CBAVD is often a **genital form of CF**.
- Two genetic causes: **CFTR mutations** and alterations in **other mesonephric duct genes**.
- **Severe mutations on both alleles → CF**; a **mild mutation on one or both → CBAVD**.
- CF carrier frequency about **1 in 20 in northern Europeans** → **test the female partner** as well.
- Mild chronic sinus or respiratory symptoms may be overlooked — the **urologist may be the first to uncover indolent CF**.
- ICSI with surgically extracted sperm is **typically effective**; one study reported **more miscarriage and stillbirth** with sperm from CFTR-mutation CBAVD men.
- **Genetic counselling** (urologist or clinical geneticist) using both partners' CFTR results explains offspring CF risk and carrier implications.

Varicocele

- Imaging is not recommended for **screening**; diagnosis is clinical.
- Most men with varicocele **have sperm**, and female-partner confounders make pregnancy effects intractable to study — yet **substantial evidence links varicocele to spermatogenic dysfunction**.
- Anatomy: the **left internal spermatic vein enters the left renal vein 8–10 cm higher** than the right enters the vena cava → taller hydrostatic column → **left valve incompetence** → varicoceles **commoner on the left**.
- With sensitive techniques (contact thermography, Doppler, venography) **most are bilateral**; clinical significance of these is uncertain.
- **Isolated right varicocele** is rare → consider **renal pathology such as tumour**, especially if of **abrupt onset**.
- Origin: **venous valvular incompetence**; **familial** clustering in first-degree relatives suggests genetic predisposition.
- Evidence of harm: **poorer bulk parameters**; **smaller testes with progressive atrophy**.

MECHANISMS OF VARICOCELE INJURY

Mechanism	Detail
Heat (primary)	Loss of countercurrent heat exchange in the pampiniform plexus → ↑ intratesticular temperature
Oxidative stress	↑ seminal oxidative stress → DNA fragmentation
Hypoperfusion and stasis	Hypoxia and toxin accumulation
Apoptosis	Apoptosis-associated microRNA raised in seminal fluid
Other	Aneuploidy predisposition; intracellular metabolic and ionic change

- **Repair improves** semen parameters, pregnancy and live birth rates, and sperm DNA fragmentation in most infertile men with **clinical** varicocele; in couples using assisted reproduction it may permit **less invasive treatment**.
- **Do not repair a non-palpable varicocele detected only on ultrasound** — no benefit for semen or pregnancy. Varicocelectomy: Chapter 70.

Concept — varicocele is common in fertile men yet over-represented in infertile men, so it is a **risk factor rather than a sentence**: repair is reserved for palpable varicoceles with abnormal semen, where heat and oxidative injury are demonstrable and correction helps.

Ejaculatory Dysfunction

- Anatomical, functional or neuropathic → **absent emission, resistance or misdirection**. Three clinical categories: **EDO, retrograde ejaculation, anejaculation** (treatment of the latter two: Chapters 70 and 74).

Ejaculatory Ductal Obstruction

- Anatomy: ejaculatory ducts are **intraprostatic extensions of the seminal vesicles without their musculature** — simple conduits. The vasal **ampulla joins medially at an acute angle**; the duct ends angled at the **verumontanum**, whose two longitudinal muscle layers extend into the urethra.
- EDO causes **<5% of azoospermia**. **Idiopathic EDO carries CFTR mutations in about half** → offer genetic evaluation.
- Causes: **infection, inflammation, prior surgery, compression by congenital cysts**, anywhere along the intraprostatic course.
- Evaluate when **semen volume <1.0 mL** (investigations: Imaging). **Partial EDO**: severe oligospermia or azoospermia, ↓ motility, ↓ volume.

Treatment: clinically significant EDO amenable to surgery → **transurethral resection of the ejaculatory ducts (TURED)** → depth guided by **synchronous TRUS** to avoid **rectal injury** → outcomes: relief of **painful ejaculation and haemospermia**, ↑ volume, improved semen quality (Chapter 70).

Structural Sperm Abnormalities

- Most sperm in fertile men are eccentrically shaped, so ordinary variation is hard to quantify clinically; certain **rare discrete defects** have overt consequences.

COMPARISON — DISCRETE STRUCTURAL SPERM DEFECTS

Defect	Morphology	Genetics	Aneuploidy	Treatment
Globozoospermia	No acrosomal cap → round heads	SPATA16 (3q26.32), PICK1 (22q12.3–q13.2), DPY19L2 (12q14.2); SPATA16 and PICK1 localise to proacrosomal granules	Debated	IVF with ICSI (no acrosome reaction possible); assisted oocyte activation proposed

Defect	Morphology	Genetics	Aneuploidy	Treatment
Macrocephaly	Large heads, multiple tails, abnormal acrosomes	Homozygous 1-bp deletion in AURKC (meiotic segregation, cytokinesis)	Very high	ICSI not recommended
Pinhead sperm	Tails only; heads absorbed	Basal plate fails to attach opposite the acrosome	—	—
Primary ciliary dyskinesia (PCD)	Immotile (to completely static), defective morphology	Autosomal recessive (occasionally X-linked); DNAH5 and DNAI1 account for a large share	—	ICSI; ejaculated, epididymal or testicular sperm — immotile testicular sperm fertilise better

Primary Ciliary Dyskinesia

- Disrupted **microtubular architecture of cilia**; the sperm tail shares it → **immotile sperm, congenital heart disease, chronic respiratory and ear–nose–throat infection, laterality defects**.
- About **1 in 15,000–30,000 live births**.

KEY POINTS: DIAGNOSES AND THERAPIES

- **Klinefelter syndrome (47,XXY)** is the commonest identified genetic cause of male infertility; **body morphology cannot reliably exclude it**.
- A man with **CBAVD** needs **CFTR assessment of himself and his female partner** to define offspring CF risk.
- Severe hypogonadotropic hypogonadism may be associated with **anosmia** and is treated with **gonadotropin replacement**; milder forms are commoner and may respond to **antioestrogens or aromatase inhibitors**.

Concept — in monomorphic sperm defects the **ultrastructure dictates the therapy**: ICSI bypasses an absent acrosome or a paralysed axoneme, but cannot correct the aneuploidy that accompanies macrocephalic multiflagellate sperm.